explore_licenses_20261101

November 1, 2026

[1]:
```
import polars as pl

from tool_extraction import artifacts, artifact_siblings, repos, mining_runs
```

0.1 License analysis

[ ]:
```
licenses = (
    repos.select(
        pl.col("license")
        .str.strip_chars()
        .str.to_lowercase()
        .fill_null("unlicense")
        .replace("", "unlicense")
        .alias("license")
    )
    .unique()
    .sort("license")
    .head(50)
    .collect()
)

with pl.Config(tbl_rows=-1):
    display(licenses)
```

```
shape: (42, 1)

 license
 ---
 str

 0bsd
 afl-3.0
 agpl-3.0
 apache-2.0
 artistic-2.0
 blueoak-1.0.0
 bsd-2-clause
 bsd-2-clause-patent
```

```
bsd-3-clause
bsd-3-clause-clear
bsd-4-clause
bsl-1.0
cc-by-4.0
cc-by-sa-4.0
cc0-1.0
cern-ohl-s-2.0
cern-ohl-w-2.0
ecl-2.0
epl-1.0
epl-2.0
eupl-1.2
gfdl-1.3
gpl-2.0
gpl-3.0
isc
lgpl-2.1
lgpl-3.0
lppl-1.3c
mit
mit-0
mpl-2.0
ms-pl
mulanpsl-2.0
noassertion
odbl-1.0
ofl-1.1
osl-3.0
postgresql
unlicense
upl-1.0
wtfpl
zlib
```